EXAM SHIELD

Question Paper

Computing question bank: 19 subjects across computing, with multiple-choice, fill-in-the-gap and written questions

247 questions: 152 multiple choice, 57 fill in the gap and 38 written. Total 384 marks.
Edition: question paper (no answers shown).

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How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Golang

Concurrency, error handling, types and tooling in Go.

1. Which keyword starts a new goroutine?

A. go
B. spawn
C. async
D. thread

2. What is the value of a map declared with var m map[string]int before it is initialised?

A. A map with one zero entry
B. An empty map ready for writes
C. It does not compile
D. nil

3. What happens when a program writes to a nil map in Go?

A. The map is created automatically
B. The write is silently ignored
C. A compile-time error
D. A runtime panic

4. Which statement schedules a function call to run when the surrounding function returns?

A. later
B. defer
C. await
D. finally

5. How does idiomatic Go report a recoverable failure from a function?

A. By throwing an exception
B. By returning an error value as the last result
C. By setting a global errno variable
D. By calling panic in every case

6. Which built-in function creates a channel of integers?

A. channel(int)
B. new(chan int)
C. make(chan int)
D. chan.New(int)

7. What does a select statement do in Go?

A. Waits on several channel operations and runs the one that is ready
B. Chooses which goroutine the scheduler stops
C. Selects rows from a database table
D. Sorts the elements of a slice

8. Which command formats Go source code to the standard style?

A. go style
B. gopretty
C. gofmt
D. golint

9. A Go type satisfies an interface _____: it needs no implements declaration. [1 mark]

10. In Go, the zero value of an int is _____ and the zero value of a string is the _____ string. [2 marks]

11. A Go module records its path and dependency versions in a file named _____. [1 mark]

12. Explain the difference between an array and a slice in Go, including how each is passed to a function. [4 marks]

13. What is a data race? Describe how the Go race detector helps you find one. [4 marks]

Section 2: Python

Core syntax, data types and idioms of Python 3.

14. Which of these Python types is immutable?

A. dict
B. tuple
C. set
D. list

15. Which keyword defines a function in Python?

A. def
B. func
C. fn
D. function

16. What does len("hello") return?

A. 6
B. 4
C. An error
D. 5

17. What is the result of the expression 7 // 2 in Python 3?

A. 1
B. 3.5
C. 3
D. 4

18. Which line is a valid dictionary comprehension?

A. [k: v for k, v in pairs]
B. (k: v for k, v in pairs)
C. {k: v for k, v in pairs}
D. dict[k for k in pairs]

19. What does a with open(path) as fh: block guarantee?

A. The file is closed when the block exits
B. The file is locked for other processes
C. The file is deleted afterwards
D. The file is opened read-only

20. Which tool installs third-party Python packages?

A. npm
B. gem
C. composer
D. pip

21. What does print(type([])) display?

A. <class 'list'>

B. <class 'dict'>

C. <class 'array'>

D. <class 'tuple'>

22. Python handles exceptions with a try block followed by one or more _____ blocks, and code in a _____ block always runs. [2 marks]

23. The special method that initialises a new object of a Python class is named _____. [1 mark]

24. An isolated per-project Python environment is created with the standard library module _____. [1 mark]

25. Explain the Global Interpreter Lock (GIL) and its effect on multi-threaded, CPU-bound Python code. [4 marks]

26. Compare a list with a generator. When is a generator the better choice? [4 marks]

Section 3: JavaScript

Language fundamentals, types and asynchronous programming.

27. Which keyword declares a block-scoped variable that cannot be reassigned?

A. var
B. let
C. const
D. static

28. What does typeof null evaluate to?

A. "null"
B. "object"
C. "number"
D. "undefined"

29. What is the result of "5" + 3 in JavaScript?

A. "53"
B. A TypeError
C. 8
D. NaN

30. Which operator compares both value and type without coercion?

A. =>
B. ==
C. =
D. ===

31. Which method turns a JSON string into a JavaScript object?

A. JSON.toObject
B. Object.fromJSON
C. JSON.parse
D. JSON.stringify

32. Which syntax, standardised in ES2017, lets you write promise-based code that reads synchronously?

A. async / await
B. XMLHttpRequest
C. Generators with yield
D. Callbacks

33. Which array method returns a new array containing each element transformed by a function?

A. push
B. splice
C. map
D. forEach

34. When does a setTimeout callback with a delay of 0 ms run?

A. After the current call stack and pending microtasks finish
B. Immediately, before the next line of code
C. Never; a delay of 0 is ignored
D. Before any pending promise callbacks

35. In the browser, the tree of objects that represents a web page is the _____, and an element can be found by its id with document._____(id). [2 marks]

36. The runtime that executes JavaScript on servers, outside the browser, is _____. [1 mark]

37. A Promise that completes successfully is _____, and one that fails is _____. [2 marks]

38. Explain what a closure is and give one practical use case. [4 marks]

39. Describe the differences between var, let and const. [3 marks]

Section 4: PHP

Server-side PHP: syntax, forms, security and tooling.

40. Which character begins every variable name in PHP?

A. @
B. $
C. #
D. &

41. Which superglobal holds data submitted by an HTML form using the POST method?

A. $_POST
B. $_FORM
C. $_SERVER
D. $_GET

42. Which operator joins two strings in PHP?

A. . (dot)
B. ++ (increment)
C. + (plus)
D. & (ampersand)

43. Which function checks that a variable exists and is not null?

A. isset()
B. is_null()
C. empty()
D. defined()

44. What is the correct way to prevent SQL injection when using PDO?

A. Calling addslashes() on input
B. Running strip_tags() on input
C. Hashing the input with md5()
D. Prepared statements with bound parameters

45. Which tool manages PHP project dependencies?

A. npm
B. pip
C. Maven
D. Composer

46. Which function should be used to hash user passwords?

A. password_hash()
B. md5()
C. base64_encode()
D. sha1()

47. What does echo 10 <=> 5; print?

A. 1
B. 0
C. -1
D. true

48. In PHP, the superglobal that holds URL query-string parameters is _____. [1 mark]

49. The PHP function that converts an array into a JSON string is _____. [1 mark]

50. To check a login password against a hash made by password_hash(), PHP code calls _____. [1 mark]

51. Explain the difference between include and require in PHP. [3 marks]

52. What are PHP sessions? List two measures that keep session data secure. [4 marks]

Section 5: Laravel

The Laravel framework: routing, Eloquent, Blade and Artisan.

53. What is the name of Laravel's command-line tool?

A. Tinker
B. Artisan
C. Forge
D. Blade

54. Which templating engine ships with Laravel?

A. Twig
B. Smarty
C. Blade
D. Mustache

55. What is Laravel's built-in ORM called?

A. Sequelize
B. Hibernate
C. Eloquent
D. Doctrine

56. Which command creates a new migration file?

A. php artisan make:migration create_posts_table
B. php artisan db:create posts
C. php artisan migrate:new posts
D. composer make:migration posts

57. In which file are browser (web) routes normally defined?

A. app/routes.php
B. routes/web.php
C. config/routes.php
D. public/routes.php

58. What is the main purpose of middleware in Laravel?

A. Defining the database schema
B. Filtering HTTP requests entering the application
C. Compiling front-end assets
D. Rendering Blade views

59. Which Blade directive adds a CSRF token field to a form?

A. @protect
B. @secure
C. @csrf
D. @token

60. Which command applies all pending database migrations?

A. php artisan db:run
B. composer migrate
C. php artisan migrate
D. php artisan schema:apply

61. Laravel validation rules can be kept in a dedicated Form _____ class instead of the controller. [1 mark]

62. Laravel reads environment-specific settings, such as the database password, from a file named _____. [1 mark]

63. Slow work such as sending email can be pushed to background workers with Laravel _____. [1 mark]

64. Explain mass assignment and how the $fillable property protects against it. [4 marks]

65. Describe Laravel's service container and how it supports dependency injection. [4 marks]

Section 6: Computer Science

Algorithms, data structures, number systems and systems.

66. What is the time complexity of binary search on a sorted array?

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)

67. Which data structure follows Last-In, First-Out (LIFO) order?

A. Queue
B. Linked list
C. Binary search tree
D. Stack

68. What is the average-case time complexity of quicksort?

A. O(n log n)
B. O(n^2)
C. O(n)
D. O(log n)

69. What is the binary representation of decimal 10?

A. 1100
B. 0110
C. 1001
D. 1010

70. Which algorithm finds shortest paths in a graph with non-negative edge weights?

A. Depth-first search
B. Dijkstra's algorithm
C. Kruskal's algorithm
D. Prim's algorithm

71. What does a compiler do?

A. Translates source code into machine or target code
B. Executes source code one line at a time
C. Compresses files for storage
D. Translates machine code back into source code

72. What is the average time to look up a key in a hash table?

A. O(1)
B. O(n^2)
C. O(n)
D. O(log n)

73. Which of these is NOT a core operating system function?

A. Compiling user programs

B. Process scheduling

C. Memory management

D. File management

74. Breadth-first search uses a _____ to hold the nodes to visit, while depth-first search uses a _____ or recursion. [2 marks]

75. A tree in which every node has at most two children is called a _____ tree. [1 mark]

76. A problem is NP-complete if it is in NP and every problem in NP can be _____ to it in polynomial time. [1 mark]

77. Explain recursion and the role of the base case. Give a simple example. [3 marks]

78. Compare a stack and a queue, giving one real-world use for each. [4 marks]

Section 7: Information Technology

Networking, infrastructure, cloud services and IT operations.

79. Which protocol automatically assigns IP addresses to devices on a network?

A. FTP
B. DNS
C. SMTP
D. DHCP

80. What is the main job of DNS?

A. Assigning MAC addresses
B. Translating domain names into IP addresses
C. Routing email between servers
D. Encrypting web traffic

81. What is the default port for HTTPS?

A. 80
B. 443
C. 21
D. 8080

82. Which device forwards traffic between different networks?

A. Repeater
B. Router
C. Hub
D. Switch

83. What does RAID 1 provide?

A. Distributed parity across three disks
B. Striping without redundancy
C. Mirroring of data across disks
D. Data compression

84. Which of these is a cloud service model?

A. LAN (Local Area Network)
B. VPN (Virtual Private Network)
C. SaaS (Software as a Service)
D. DHCP (Dynamic Host Configuration Protocol)

85. How many bits are in an IPv4 address?

A. 32
B. 64
C. 128
D. 16

86. What does the 3-2-1 backup rule recommend?

A. 3 servers, 2 administrators and 1 user

B. Daily, weekly and monthly backups only

C. 3 copies of data, on 2 types of media, with 1 copy offsite

D. 3 disks in a RAID 2 array plus 1 spare

87. The command that shows the route packets take to a destination is _____ on Linux and tracert on Windows. [1 mark]

88. A device that keeps computers running for a short time during a power failure is a(n) _____. [1 mark]

89. The widely used framework of IT service management practices, such as incident and change management, is _____. [1 mark]

90. Explain the difference between a LAN and a WAN, with one example of each. [3 marks]

91. Describe the role of an IT help desk and how a ticketing system supports it. [4 marks]

Section 8: Information Systems

How organisations use data, systems and processes.

92. What does TPS stand for in information systems?

A. Team Performance System
B. Total Planning Service
C. Text Processing Software
D. Transaction Processing System

93. What does an ERP system do?

A. Handles only accounting
B. Hosts the company website
C. Manages only email
D. Integrates core business processes across departments

94. How does information differ from data?

A. There is no difference
B. Information is raw, unprocessed facts
C. Information is data processed into a meaningful form
D. Information refers to computer hardware

95. What is a Decision Support System (DSS) mainly used for?

A. Printing documents
B. Running payroll only
C. Helping managers analyse options and make decisions
D. Routing network traffic

96. What is the main goal of database normalisation?

A. Increasing hardware speed
B. Adding more users
C. Encrypting the database
D. Reducing data redundancy

97. Which SDLC phase identifies what users need from a new system?

A. Deployment
B. Maintenance
C. Testing
D. Requirements analysis

98. What does a CRM system focus on?

A. Managing CPU scheduling
B. Managing legal compliance
C. Managing source code
D. Managing relationships with customers

99. What does a primary key do in a relational table?

A. Encrypts the table
B. Sorts rows alphabetically
C. Uniquely identifies each row
D. Links to a table in another database only

100. A diagram showing how data moves between processes, data stores and external entities is a _____. [1 mark]

101. A system that gives senior executives summarised, high-level information is an _____ system. [1 mark]

102. Business _____ tools analyse historical data to produce dashboards and reports for decision makers. [1 mark]

103. Explain how a Management Information System (MIS) supports middle management. [4 marks]

104. Discuss two ethical issues that arise from the use of information systems. [4 marks]

Section 9: Cyber Security

Security principles, threats, cryptography and defence.

105. What do the letters of the CIA triad stand for?

A. Cipher, Identity, Access
B. Control, Inspection, Audit
C. Compliance, Integrity, Authentication
D. Confidentiality, Integrity, Availability

106. What is phishing?

A. Guessing passwords by trying every combination
B. Tricking people into revealing information through fake messages
C. Capturing packets on a network
D. Flooding a server with requests

107. Which of these is an asymmetric (public-key) algorithm?

A. AES
B. DES
C. 3DES
D. RSA

108. What is the most effective defence against SQL injection?

A. Longer passwords
B. Parameterised queries
C. Using HTTPS
D. Installing a firewall

109. What does multi-factor authentication require?

A. Two or more independent proofs of identity
B. Two or more administrator accounts
C. Two or more firewalls
D. Two or more backups

110. Which property describes a cryptographic hash function?

A. It is one-way and cannot practically be reversed
B. It compresses files losslessly
C. It can be decrypted with the right key
D. It is the same as Base64 encoding

111. A DDoS attack mainly threatens which security goal?

A. Confidentiality
B. Integrity
C. Non-repudiation
D. Availability

112. What does the principle of least privilege mean?

A. Every user is an administrator
B. All users have read-only access to everything
C. Passwords are not required
D. Users get only the access they need to do their work

113. A _____ filters network traffic according to a set of security rules. [1 mark]

114. Malware that encrypts a victim's files and demands payment for the key is called _____. [1 mark]

115. Simulating a real attack on a system, with the owner's permission, to find weaknesses is called penetration _____. [1 mark]

116. Explain the difference between symmetric and asymmetric encryption, with an example of each. [4 marks]

117. Describe the main phases of an incident response plan. [5 marks]

Section 10: Software Engineering

Processes, design principles, testing and collaboration.

118. Which agile framework organises work into fixed-length sprints?

A. Spiral
B. Scrum
C. Waterfall
D. V-model

119. What does unit testing focus on?

A. Only the user interface
B. Only the hardware
C. The smallest testable parts of a program
D. The whole system end to end

120. Which Git command records staged changes in the repository history?

A. git push
B. git commit
C. git clone
D. git status

121. In the SOLID principles, what does the S stand for?

A. Simple design
B. Static typing
C. Single responsibility
D. Secure coding

122. What is refactoring?

A. Rewriting the program in another language
B. Adding new features
C. Improving code structure without changing its behaviour
D. Deleting failing tests

123. What does CI stand for in modern development practice?

A. Code Inspection
B. Central Index
C. Compiled Interface
D. Continuous Integration

124. Which UML diagram shows how objects interact over time?

A. Sequence diagram
B. Class diagram
C. Entity-relationship diagram
D. Deployment diagram

125. What is technical debt?

A. Unpaid developer salaries
B. Loans used to buy hardware
C. Money owed for software licences
D. The future cost of shortcuts taken in code today

126. A Git _____ lets a developer work on a feature in isolation before merging it. [1 mark]

127. The design pattern that ensures a class has only one instance is the _____ pattern. [1 mark]

128. Re-running tests to check that existing features still work after a change is _____ testing. [1 mark]

129. Compare the Waterfall and Agile approaches to software development. [5 marks]

130. Explain the purpose of code reviews and two benefits they bring to a team. [3 marks]

Section 11: DevOps Engineering

Containers, orchestration, automation and observability.

131. In Docker, what is a container?

A. A hypervisor for virtual machines
B. A registry that stores images
C. A running instance of an image
D. A script that builds an image

132. Which file describes how to build a Docker image?

A. Dockerfile
B. Makefile
C. Jenkinsfile
D. package.json

133. What is the smallest deployable unit in Kubernetes?

A. Cluster
B. Pod
C. Namespace
D. Node

134. Which tool is used for Infrastructure as Code?

A. Terraform
B. Wireshark
C. Postman
D. Photoshop

135. What is a blue-green deployment?

A. A debugging technique
B. Running two identical environments and switching traffic between them
C. A code formatting standard
D. A colour scheme for dashboards

136. Which tool collects and stores time-series metrics for monitoring?

A. Webpack
B. Babel
C. Prometheus
D. Figma

137. In a CI/CD pipeline, what normally happens in the test stage?

A. Servers are physically installed
B. Automated tests run against the new build
C. Old builds are archived
D. The release is announced to customers

138. Which command lists the running Docker containers?

A. docker show
B. docker ps
C. docker ls-run
D. docker images

139. Releasing a new version to a small share of users before everyone else is a _____ deployment. [1 mark]

140. Ansible, Puppet and Chef are _____ management tools. [1 mark]

141. Keeping the desired state of infrastructure in Git and applying changes automatically from it is called _____. [1 mark]

142. Explain what a CI/CD pipeline is and describe three benefits it brings. [5 marks]

143. Compare containers with virtual machines in terms of isolation, size and start-up time. [4 marks]

Section 12: Computer Networks

Layers, addressing, routing and core Internet protocols.

144. Which OSI layer is responsible for routing packets between networks?

A. Data link
B. Session
C. Transport
D. Network

145. Which protocol finds the MAC address that belongs to an IPv4 address on a local network?

A. ICMP
B. DNS
C. ARP
D. NAT

146. How many usable host addresses does a /24 IPv4 subnet provide?

A. 254
B. 255
C. 256
D. 252

147. Which transport protocol provides reliable, ordered delivery of data?

A. UDP
B. IP
C. ICMP
D. TCP

148. Which protocol transfers email between mail servers?

A. SMTP
B. POP3
C. SNMP
D. IMAP

149. Which of these is a link-state routing protocol?

A. RIPng
B. RIP
C. OSPF
D. BGP

150. Which device forwards frames inside a LAN using MAC addresses?

A. Router
B. Repeater
C. Hub
D. Switch

151. What does NAT mainly allow?

A. Encryption of all traffic

B. Automatic IP address assignment

C. Translation of names into addresses

D. Many private hosts to share one public IP address

152. The ping utility sends messages using the _____ protocol. [1 mark]

153. DNS queries normally use port _____. [1 mark]

154. An IPv6 address is _____ bits long, compared with _____ bits for IPv4. [2 marks]

155. Explain the TCP three-way handshake and why a connection needs it. [5 marks]

156. Compare IPv4 and IPv6 in terms of address size, header design and the need for NAT. [5 marks]

Section 13: Internet of Things and Embedded Systems

Sensors, microcontrollers, IoT protocols and edge computing.

157. Which lightweight publish/subscribe protocol is widely used in IoT?

A. SNMP
B. FTP
C. SMTP
D. MQTT

158. In MQTT, which component receives messages from publishers and passes them to subscribers?

A. Client
B. Gateway
C. Router
D. Broker

159. Which wireless technology is designed for low-power IoT links over several kilometres?

A. Bluetooth Classic
B. Wi-Fi 6
C. NFC
D. LoRaWAN

160. A microcontroller turns an analog sensor voltage into a number using a(n):

A. UART
B. DAC
C. ADC
D. PWM output

161. Which technique varies the duty cycle of a signal to control motor speed or LED brightness?

A. I2C
B. SPI
C. ADC
D. PWM

162. Which two-wire serial bus uses SDA and SCL lines to connect several devices?

A. I2C
B. UART
C. SPI
D. Ethernet

163. An embedded designer chooses a real-time operating system (RTOS) mainly for:

A. A graphical desktop
B. Built-in cloud storage
C. Deterministic task scheduling and timing
D. Unlimited memory

164. Why is CoAP suited to constrained IoT devices?

A. It only uses XML messages
B. It needs a persistent TCP connection
C. It runs over UDP with a compact binary header
D. It requires a web browser

165. Processing data close to where it is produced, instead of in a distant data centre, is called _____ computing. [1 mark]

166. MQTT messages are published to named _____, and the highest MQTT QoS level is _____. [2 marks]

167. A _____ timer automatically resets a microcontroller if its software stops responding. [1 mark]

168. Describe a layered IoT architecture (devices, gateway, cloud, application) for a smart-farming system, and explain where data should be processed and why. [5 marks]

169. Discuss three security challenges specific to IoT devices and one mitigation for each. [5 marks]

Section 14: Artificial Intelligence and Machine Learning

Learning paradigms, model evaluation, neural networks and responsible AI.

170. Training a model on labelled input-output examples is called:

A. Clustering
B. Unsupervised learning
C. Supervised learning
D. Reinforcement learning

171. Which of these is an unsupervised clustering algorithm?

A. Logistic regression
B. Linear regression
C. k-means
D. Naive Bayes

172. Which metric is the harmonic mean of precision and recall?

A. Accuracy
B. Mean squared error
C. ROC AUC
D. F1 score

173. Which technique reduces overfitting by randomly switching off neurons during training?

A. Padding
B. Momentum
C. Pooling
D. Dropout

174. Which neural network architecture underlies modern large language models?

A. Transformer
B. Single perceptron
C. Convolutional neural network
D. Hopfield network

175. In reinforcement learning, what does an agent try to maximise?

A. Number of labels
B. Training loss
C. Learning rate
D. Cumulative reward

176. Splitting data into k parts and training k times, holding out a different part each time, is called:

A. k-fold cross-validation
B. Grid search
C. Bagging
D. Bootstrapping

177. Which activation function outputs max(0, x)?

A. Tanh
B. Softmax
C. ReLU
D. Sigmoid

178. Neural networks compute gradients by applying the chain rule backwards through the layers, an algorithm called _____. [1 mark]

179. Data used to tune hyperparameters is the _____ set, and data kept aside for the final estimate of performance is the _____ set. [2 marks]

180. A _____ matrix tabulates true positives, false positives, true negatives and false negatives. [1 mark]

181. Explain the bias-variance trade-off and how it relates to underfitting and overfitting. [5 marks]

182. Discuss two sources of unfair bias in machine-learning systems and how responsible-AI practice reduces them. [5 marks]

Section 15: Data Science and Databases

SQL, database design, transactions, warehousing and analytics.

183. Which SQL command removes rows from a table?

A. ERASE
B. REMOVE
C. DROP
D. DELETE

184. Which SQL keyword removes duplicate rows from a query result?

A. UNIQUE ROWS
B. SINGLE
C. ONLY
D. DISTINCT

185. Which normal form removes transitive dependencies?

A. Second normal form
B. First normal form
C. Third normal form
D. Unnormalised form

186. Which join returns every row of the left table plus matching rows from the right table?

A. CROSS JOIN
B. INNER JOIN
C. SELF JOIN
D. LEFT JOIN

187. Which kind of database stores records as flexible JSON-like documents?

A. Spreadsheet
B. Document database
C. Relational database
D. Flat file

188. In data engineering, ETL stands for:

A. Evaluate, Test, Launch
B. Encrypt, Transfer, Log
C. Export, Train, Learn
D. Extract, Transform, Load

189. Which measure of central tendency is least affected by outliers?

A. Median
B. Range
C. Mean
D. Variance

190. An index on a column mainly speeds up:

A. Searching and filtering on that column
B. Taking backups
C. Changing the schema
D. Inserting new rows

191. In SQL, rows that share a value are combined with GROUP _____, and the resulting groups are filtered with _____. [2 marks]

192. A foreign key enforces _____ integrity between two tables. [1 mark]

193. In a data-warehouse star schema, a central _____ table is surrounded by dimension tables. [1 mark]

194. Explain the ACID properties of a transaction, with an example of what goes wrong without each. [5 marks]

195. Describe the stages of a data-science project from problem definition to deployment, and the role of data cleaning. [5 marks]

Section 16: Cloud Computing

Service and deployment models, scaling, storage and shared responsibility.

196. Which cloud service model gives customers virtual machines, storage and networks to manage themselves?

A. IaaS
B. PaaS
C. FaaS
D. SaaS

197. Gmail and Microsoft 365 are examples of which service model?

A. PaaS
B. IaaS
C. On-premises software
D. SaaS

198. Serverless functions are mainly billed by:

A. Servers rented per month
B. A fixed annual licence
C. Disk size only
D. Invocations and execution time

199. Automatically adding instances when load rises is called:

A. Auto-scaling
B. Vertical scaling
C. Load shedding
D. Caching

200. In IaaS, who secures the physical data centre under the shared responsibility model?

A. Both equally
B. Nobody
C. The customer
D. The cloud provider

201. Which tool orchestrates containers across a cluster of machines?

A. Jenkins
B. Kubernetes
C. Grafana
D. Git

202. A deployment combining a private data centre with a public cloud is a:

A. Community cloud
B. Single-tenant cloud
C. Edge cloud
D. Hybrid cloud

203. Object storage services such as Amazon S3 store data as:

A. Objects in buckets
B. Files on tape
C. Blocks on a mounted disk
D. Rows in tables

204. Paying only for the resources actually consumed is called _____-as-you-go pricing. [1 mark]

205. Incoming traffic is spread across several servers by a load _____. [1 mark]

206. Running code without provisioning or managing servers is known as _____ computing. [1 mark]

207. Compare IaaS, PaaS and SaaS, giving an example of each and the customer's responsibilities in each. [5 marks]

208. Discuss vendor lock-in and data sovereignty risks when a university moves to the cloud, and how to reduce them. [5 marks]

Section 17: Computer Architecture and Operating Systems

Processors, memory hierarchy, processes, scheduling and concurrency.

209. Which CPU component performs arithmetic and logical operations?

A. Control unit
B. ALU
C. Program counter
D. Cache

210. Which of these memories is fastest?

A. Solid-state drive
B. Main memory (RAM)
C. L3 cache
D. CPU registers

211. Deadlock needs mutual exclusion, hold and wait, no preemption and:

A. Starvation
B. Circular wait
C. Paging
D. Thrashing

212. Which scheduling algorithm can starve long jobs?

A. Round robin
B. Fair-share scheduling
C. First-come, first-served
D. Shortest job first

213. Virtual memory lets a process:

A. Run without a CPU
B. Use more memory than is physically installed
C. Bypass the operating system
D. Share CPU registers with other processes

214. Code running in kernel mode differs from user mode because it:

A. Has unrestricted access to the hardware
B. Cannot handle interrupts
C. Has no memory of its own
D. Can only run applications

215. With byte addressing, a 32-bit address can reach at most:

A. 16 GB
B. 2 GB
C. 64 KB
D. 4 GB

216. Which number system uses base 16?

A. Decimal
B. Hexadecimal
C. Octal
D. Binary

217. The hexadecimal number FF equals _____ in decimal. [1 mark]

218. When the CPU stops one process and starts another, the operating system performs a context _____. [1 mark]

219. Excessive paging that keeps the CPU waiting for the disk is called _____. [1 mark]

220. Explain the fetch-decode-execute cycle and how pipelining improves throughput. [5 marks]

221. Compare processes and threads, and explain how a race condition occurs and how it can be prevented. [5 marks]

Section 18: Distributed Systems and Blockchain

Consistency, consensus, replication, fault tolerance and blockchains.

222. By the CAP theorem, during a network partition a distributed system must choose between:

A. Latency and storage
B. Consistency and availability
C. Security and usability
D. Speed and cost

223. Which algorithm lets replicated servers agree on a shared log?

A. Dijkstra's algorithm
B. Huffman coding
C. RSA
D. Raft

224. Bitcoin miners compete to add blocks using:

A. Proof of stake
B. Proof of work
C. Round-robin turns
D. Two-phase commit

225. A smart contract is:

A. An encrypted email
B. A legal document signed online
C. Code stored and executed on a blockchain
D. A cloud service-level agreement

226. Replicating data across several nodes mainly improves:

A. Compile speed
B. Availability and fault tolerance
C. Screen resolution
D. Code readability

227. Two-phase commit is used to:

A. Make a transaction atomic across several nodes
B. Collect garbage memory
C. Hash passwords
D. Route network packets

228. Eventual consistency means that:

A. Concurrent writes are rejected
B. Data is never replicated
C. All replicas converge if no new updates arrive
D. Every read returns the latest write

229. What makes past blocks in a blockchain tamper-evident?

A. Each block contains the hash of the previous block

B. Blocks are compressed

C. Blocks are numbered in order

D. Blocks are kept in RAM

230. A system that keeps working correctly when some of its components fail is fault _____. [1 mark]

231. In Raft, one server is elected _____ and the others act as followers. [1 mark]

232. Splitting a large database across machines by key ranges or hashes is called _____. [1 mark]

233. Explain why clocks cannot be perfectly synchronised in a distributed system, and how logical clocks order events. [5 marks]

234. Evaluate whether a blockchain suits storing university certificates, compared with a conventional database. [5 marks]

Section 19: Human-Computer Interaction and Mobile Development

Usability, accessibility, interaction design and mobile app platforms.

235. Which usability heuristic says users should always know what the system is doing?

A. Flexibility and efficiency of use
B. Visibility of system status
C. Error prevention
D. Aesthetic and minimalist design

236. The WCAG guidelines are concerned with:

A. Encryption strength
B. Web accessibility
C. Network routing
D. Database tuning

237. A low-fidelity sketch of a screen's layout is called a:

A. Sprint
B. Compiler
C. Wireframe
D. Schema

238. Which language does Google recommend for modern Android apps?

A. Objective-C
B. Swift
C. Kotlin
D. Perl

239. Which framework builds iOS and Android apps from one Dart codebase?

A. Laravel
B. Django
C. React
D. Flutter

240. Usability testing mainly measures:

A. Battery chemistry
B. Server uptime
C. How easily real users complete tasks
D. Code coverage

241. Responsive web design adapts a page's layout using:

A. CSS media queries
B. Firewall rules
C. Cron jobs
D. Database triggers

242. Fitts's law says the time to reach a target depends on:

A. Its distance and size

B. Network speed

C. Screen brightness

D. Its colour

243. Screen readers announce images using the text in the HTML _____ attribute. [1 mark]

244. A clickable mock-up used to test a design with users before it is built is a _____. [1 mark]

245. Apple's main programming language for iOS apps is _____. [1 mark]

246. Describe a user-centred design process for a student exam app, including how users are involved at each stage. [5 marks]

247. Compare native and cross-platform mobile development in terms of performance, cost and access to device features. [5 marks]